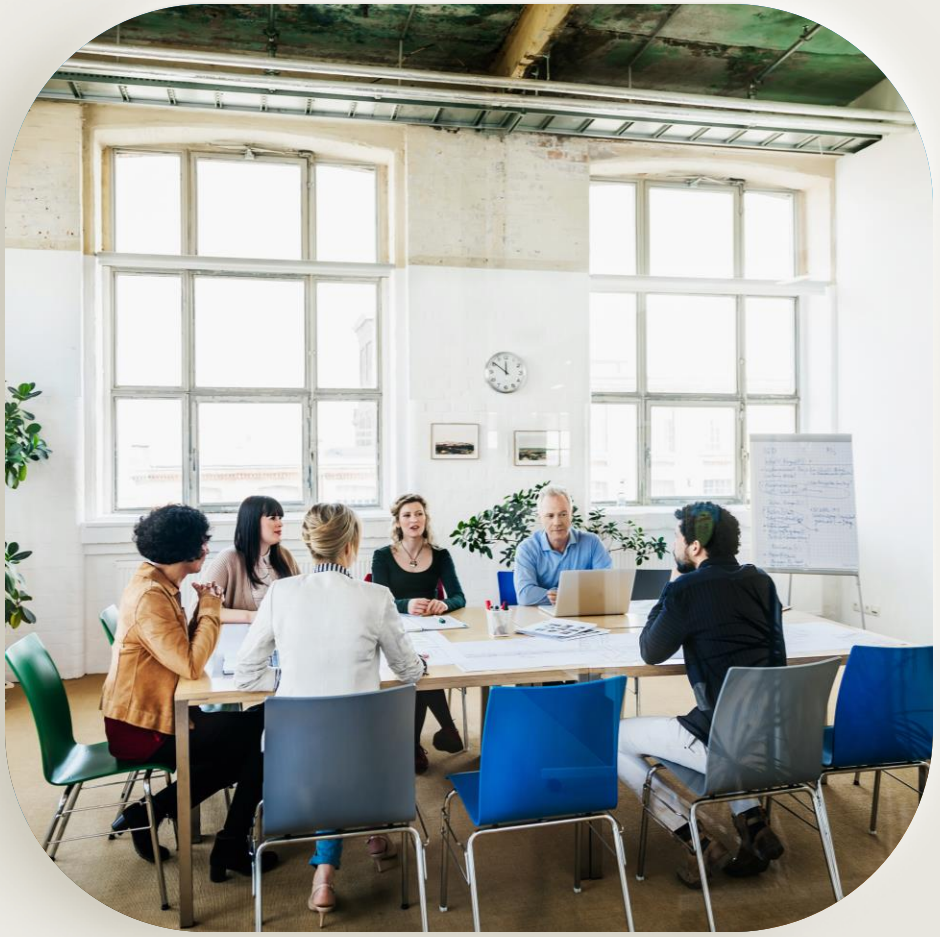
Microsoft

Microsoft Partner Credential Ready

AI-103
Exam Prep



<Presenter>

<Date>

Technofocus

Course Plan and Learning Objectives



- Plan and manage an Azure AI solution
- Implement generative AI and agentic solutions
- Implement computer vision solutions
- Implement text analysis solutions
- Implement information extraction solutions

Exam Scoring

700

MINIMUM PASSING SCORE

Each Exam has cut score

- Score ranges from 0-1000
- Passing score is 700(scaled score not 70%)



Key Strategies

Tips and Tricks

- No points deducted for wrong answers
- Some questions can not be revisited
- Answer every question

AI-103 exam at glance

Skills measured — weights from the official study guide

M1	**Plan and manage an Azure AI solution** Foundry services, models, deployment, responsible AI	Weight 25–30%
M2	**Implement generative AI and agentic solutions** Generative apps, agents, tools, optimization — the largest domain	Weight 30–35%
M3	**Implement text analysis solutions** Language model text analysis, speech, translation	Weight 10–15%
M4	**Implement computer vision solutions** Image/video generation, multimodal understanding	Weight 10–15%
M5	**Implement information extraction solutions** Retrieval and grounding pipelines, document extraction	Weight 10–15%

Question types you may see

01

Multiple choice

One correct answer from four or more

02

Multiple response

“Choose two / three” — each correct choice may score

03

Drag and drop / build list

Match items or put steps in order

04

Hot area / code completion

Pick values in a code snippet or UI

05

Yes / No series

Same scenario, different proposed solutions

06

Case study

Long scenario with several questions

Checklist

Online proctored	Test centre
System test the day before	Arrive 15–30 minutes early
Clear desk, one monitor, webcam on	Two forms of ID if required
Check in 30 minutes early with ID	Phone, bag to be kept in provided locker

Manage time

About 1.5–2 minutes per question. Leave 10 minutes at the end for flagged items.

Answer what you know quickly; flag anything over ~90 seconds

Budget time up front for Case Study — they take longer per question

Return to flagged items with fresh eyes

Final check – Do not leave out any question.

Microsoft

Plan and manage an Azure AI solution



Plan and manage an Azure AI solution

What the exam measures

- Choose the appropriate Foundry services for generative AI and agents
- Set up AI solutions in Foundry
- Manage, monitor, and secure AI systems
- Implement responsible AI across generative AI and agentic systems

25–30%

of exam questions

Plan and manage key concepts

What the exam measures

Choose the appropriate Foundry services for generative AI and agents

Foundry Tools a set of out-of-the-box prebuilt APIs and models

Model choice start with the relevant scenario leaderboard

Set up AI solutions in Foundry

Endpoints Each project has both a Project endpoint and an Azure OpenAI endpoint.

Deployment type Global Standard deployments in Foundry resources should be used whenever possible

Manage, monitor, and secure AI systems

Authentication production applications should use Microsoft Entra ID authentication

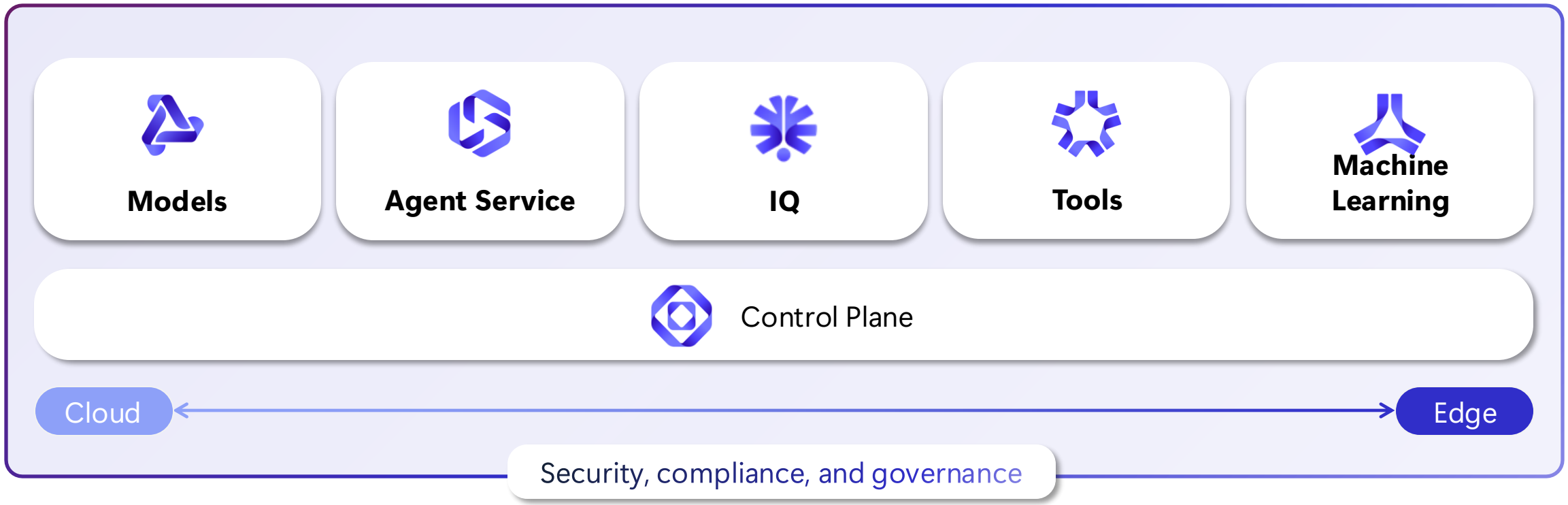
Implement responsible AI across generative AI and agentic systems

Mitigation a layered approach

Groundedness Determines whether responses are based on provided context rather than speculation.

Microsoft Foundry

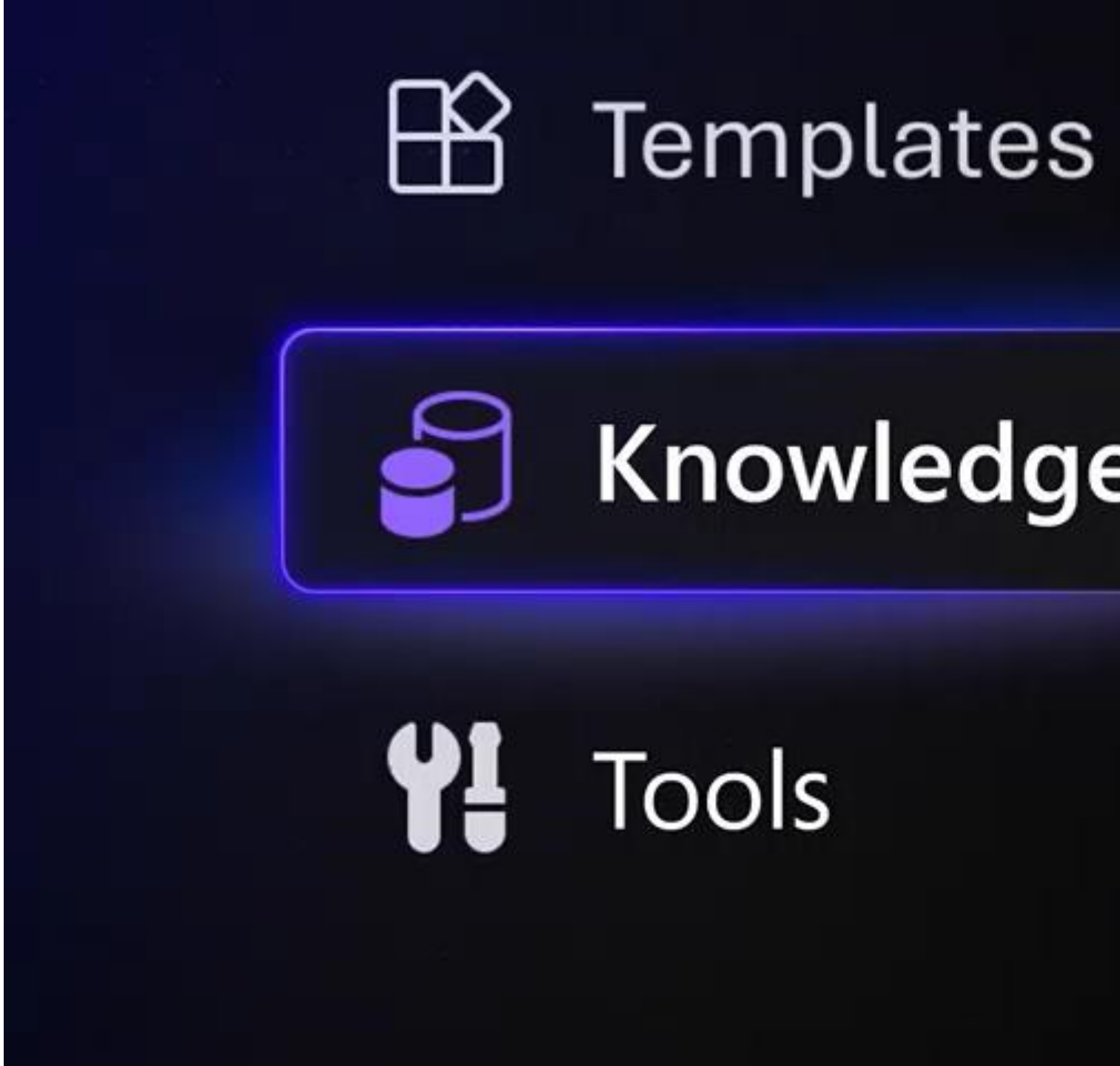
The AI app and agent factory



Foundry Tools

Empower agents to act on your business context

- Accelerated time-to-value
- Seamless integration to AI apps and agents
- Launch faster, with confidence



Foundry Models

Every model.
Every use case.
One platform.

Avoid model lock-in:

- Flexible model selection
- Curated for enterprise
- Explore, compare, and swap models quickly

Leverage the widest selection of models on any cloud

11,000+ foundational, open, reasoning, multimodal, and industry-specific models

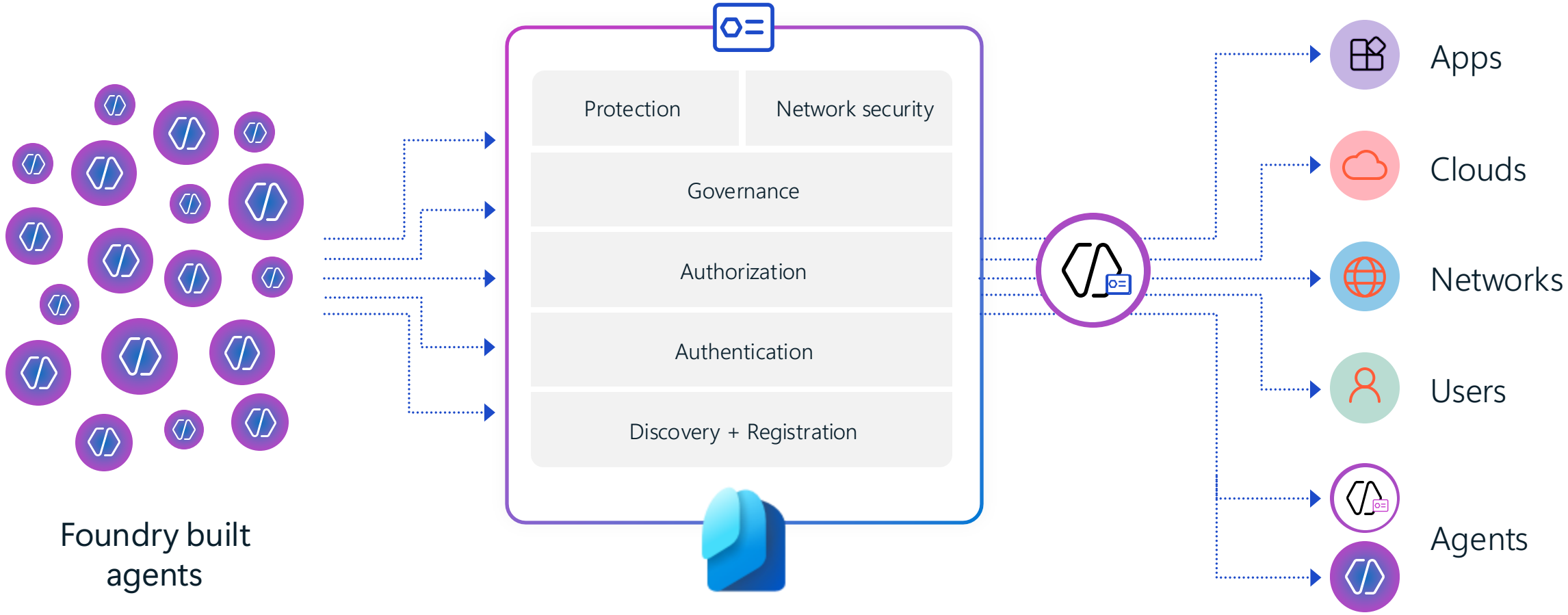
Azure OpenAI
Claude in Microsoft Foundry
Azure DeepSeek
Azure Mistral
Azure Grok
Azure Cohere
Azure Black Forest Labs
Azure Llama

and models from

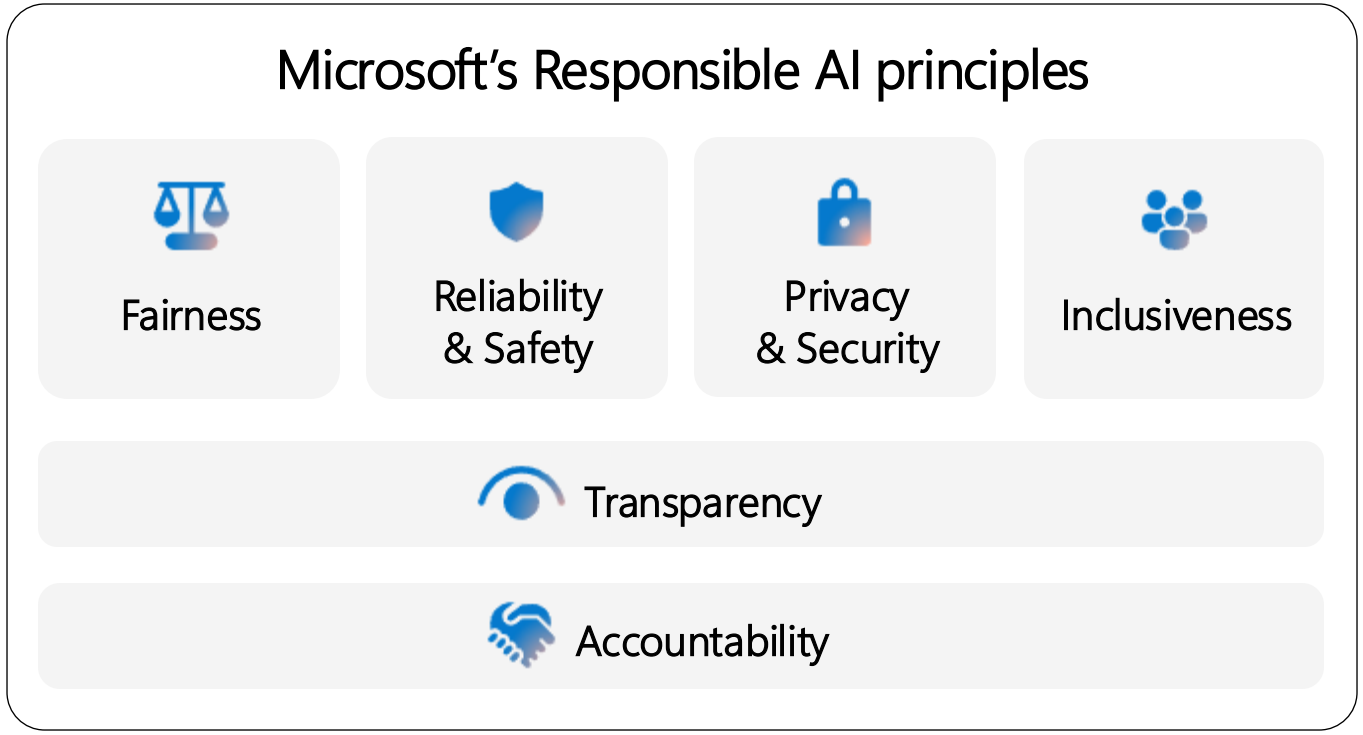
Hugging Face
NVIDIA
Nixtla
Databricks
SIGHT MACHINE
saifr
BAYER
Rockwell Automation
Paige

Microsoft Entra Agent ID

Secure access for AI agents – just as you do for employees – with enterprise-grade access management, protection, and governance of agent identities.



Responsible by design



Choose and deploy

Exam Tip

Topic	Remember
Resource vs project	Resource = compute, model delivery, networking, billing · Project = isolated workspace for agents, files, evaluations
Model catalog	Sold directly by Azure → billed via your subscription · Partner & community → purchased from the provider
Model choice	LLM for complex reasoning; SLM / Foundry Tool for narrow high-volume tasks; embedding model for vectors
Deployment types	Global Standard (default) · Data Zone (EU/US) · Provisioned PTU (steady load) · Batch (offline, 24 h)
429 errors	Rate limit — backoff honouring retry-after, more TPM, spread load
Deployment name	Code calls the deployment name, not the model name
Responsible AI	Map → Measure → Mitigate → Manage; mitigate in layers: model, safety system, system message, UX

Sample question 1

Fabrikam's procurement policy requires every production model to be billed through the company's existing Azure subscription, with no separate purchase from the model provider. You are filtering the Foundry model catalog for a summarisation model.

Which models should you shortlist?

A Foundry Models from partners and community

B Foundry Models sold directly by Azure

C Any model shown under the "fine-tuning support" filter

D Only models that support the Global Batch deployment type

Sample question 2

Three developer teams at Contoso are each building a separate agent. Each team needs an isolated workspace for its agents, files and evaluations. IT wants a single Azure resource through which networking, security and billing are managed.

What should you create?

A One Foundry resource for each team, each with a single project

B One Foundry project shared by all three teams, with a naming convention per team

C One Azure AI Search service for each team, connected to a shared project

D One Foundry resource with a separate Foundry project for each team

Sample question 3

A nightly job must generate descriptions for five million products. Results are needed by the next morning. You must minimize cost. Which deployment type should you use?

A Global Batch

B Provisioned (PTU)

C Standard (regional)

D Data Zone Standard

Sample question 4

A web app on Azure App Service must call a Foundry model without storing any keys or secrets. Which TWO actions should you take?

Choose TWO. Each correct answer is a partial solution.

A Store the API key in App Service application settings

B Enable a managed identity on the App Service

C Rotate the API key every 30 days

D Assign the identity an appropriate Azure AI role on the Foundry project or resource

Sample question 5

Tailwind Traders' retail chatbot runs on a Standard model deployment. During a two-hour flash sale, traffic triples and users intermittently get HTTP 429 errors. The rest of the year, traffic is low and spiky. You must keep the chatbot responsive without a long-term capacity commitment.

What should you do FIRST in the application?

A Disable the content filter

B Increase the temperature

C Purchase provisioned throughput (PTU) for the full year

D Retry with exponential backoff, honouring the retry-after header

Microsoft

Implement generative AI and agentic solutions



Implement generative AI and agentic solutions

What the exam measures

- Build generative applications by using Foundry
- Build agents by using Foundry
- Optimize and operationalize generative AI systems
- Choose appropriate memory, tool, and knowledge integration services for agent solutions

30–35%

of exam questions

Generative AI and agents: key concepts

What the exam measures

Build generative applications by using Foundry

Responses API the recommended approach for generating AI responses in Microsoft Foundry applications

previous_response_id links responses together, maintaining conversation context

Tools By default, the model chooses when to use a tool (and which one), based on the prompt.

Optimize and operationalize generative AI systems

Prompt engineering the foundation that supports both directions

RAG Add RAG if accuracy matters

Fine-tuning Add fine-tuning if consistency matters

Generative AI and agents: key concepts

What the exam measures

Build agents by using Foundry

Custom tools OpenAPI specified tools, Azure Functions, and function calling

OpenAPI auth anonymous, API key, and managed identity

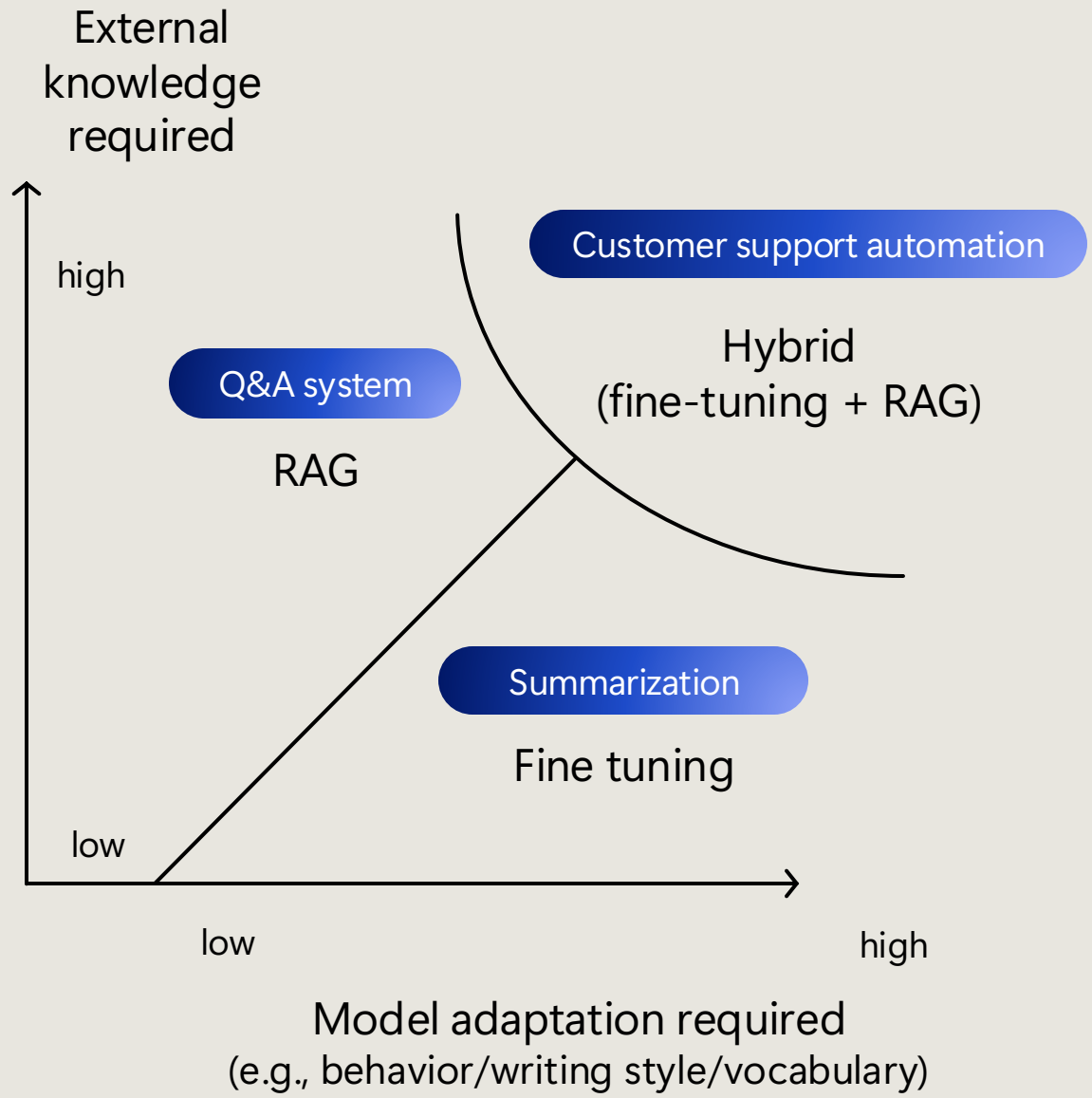
Handoff orchestration Dynamically transfer control between agents based on context or rules.

Choose appropriate memory, tool, and knowledge integration services for agent solutions

Foundry IQ a managed knowledge platform for AI agents built on Azure AI Search

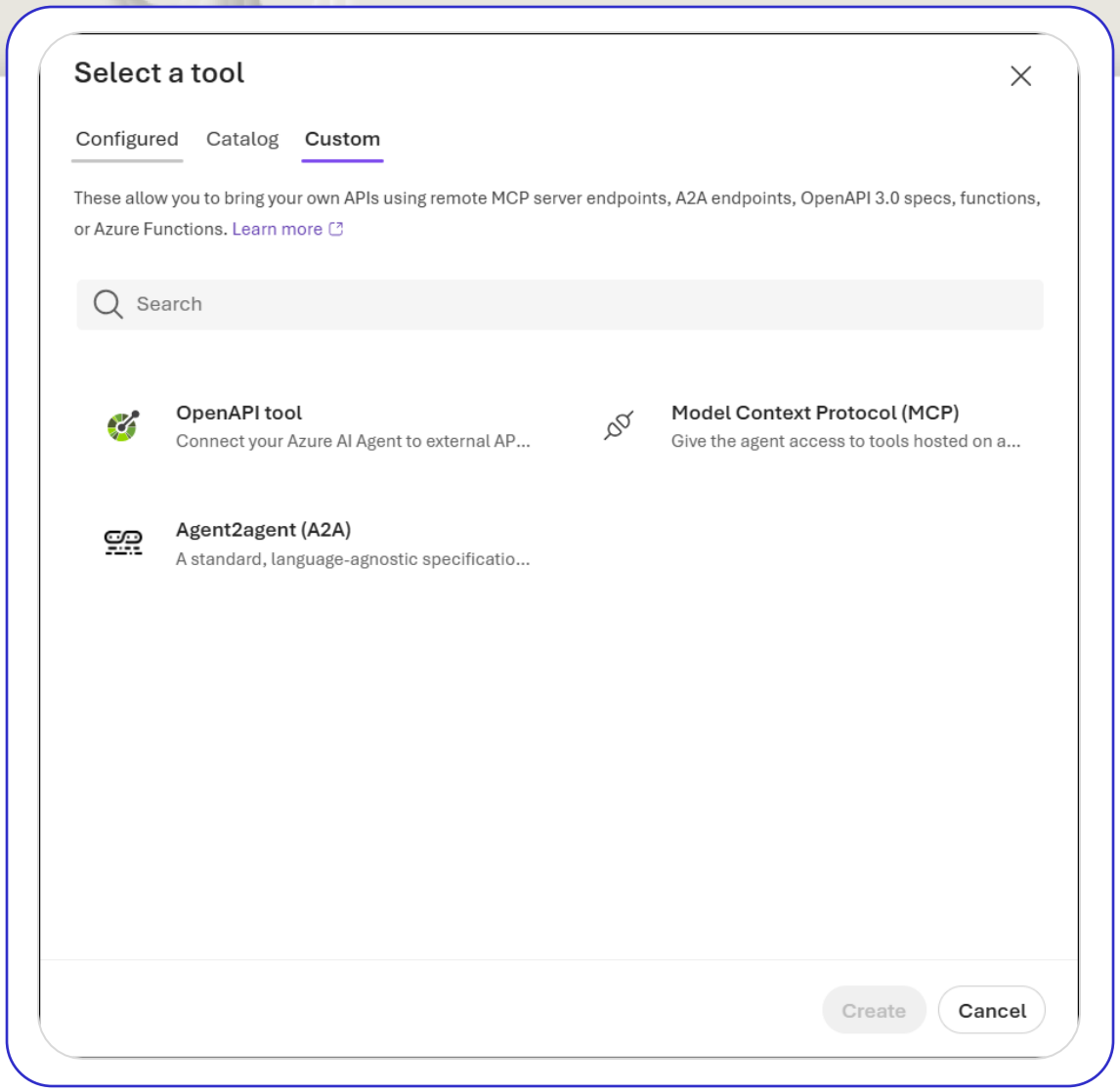
Knowledge bases organize information by business domain rather than technical storage location

When to use RAG vs. fine-tuning?

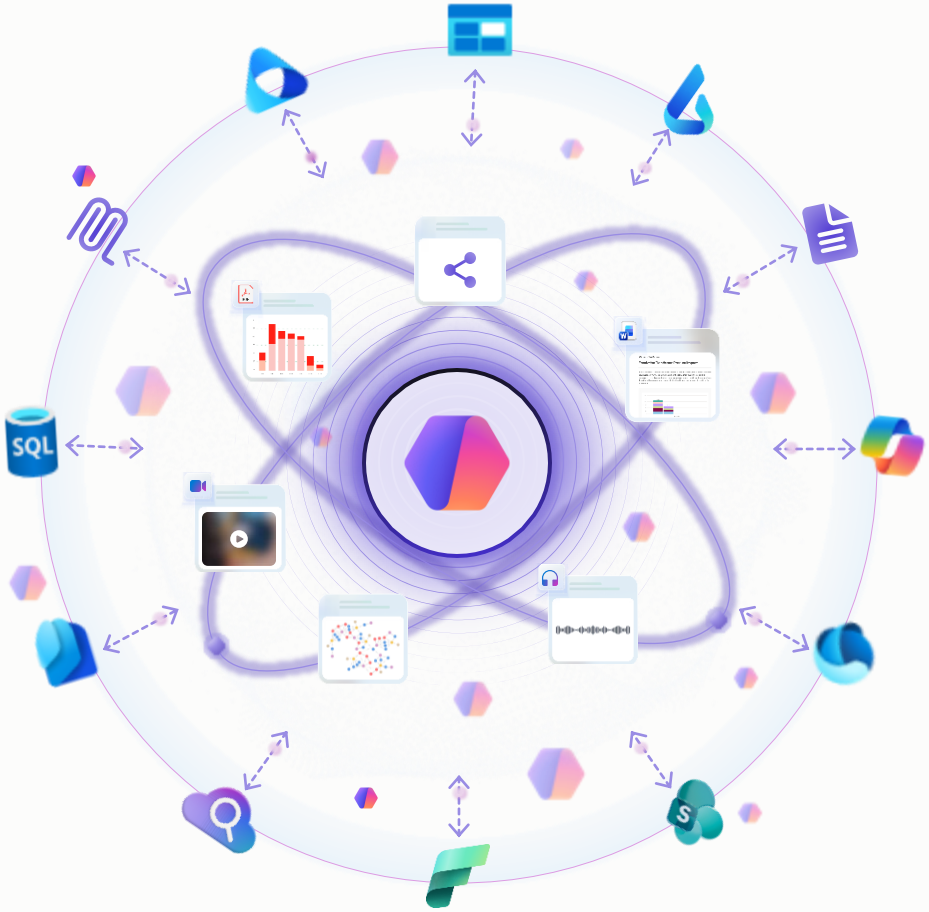


Custom tools

Tool	Description
Model Context Protocol (preview)	Give the agent access to tools hosted on an existing MCP endpoint
OpenAPI 3.0 Specified tool	Connect your Foundry agents to external APIs using functions with an OpenAPI 3.0 specification.
Agen2Agent tool (preview)	Connect your Foundry agents to another A2A compatible agents



Foundry IQ



Activate knowledge from everywhere

Faster agent delivery with point and click knowledge bases, reusable across agents.

Context without blind spots

Unlock better results with agentic retrieval that finds the relevant data automatically

Enterprise-grade security, with identity and policy built-in

Every answer respects your organization's security, identity, and compliance policies by default

Foundry IQ

Unlocking knowledge for your agents

Reusable knowledge bases

Indexed & remote knowledge sources

Agentic RAG for superior context

Enterprise-grade security

Point-and-click data transformation & embedding

Sources



Integrations



Sample question 6

You are creating a travel-booking agent in Foundry. It must always confirm trip dates and budget before booking, and prefer direct flights when the price difference is under 20%. Where should you define this behaviour?

A In an OpenAPI tool definition

B In the content filter configuration of the model deployment

C In the agent's instructions

D In the vector store used by the file search tool

Sample question 7

Litware is building a chat app with the Responses API. Employees ask about today's public exchange rates and about rules in the company's travel policy PDFs, which have already been uploaded to a vector store.
Which TWO tools should you add to the request c*hoose TWO. Each correct answer is a partial solution.*

A The code_interpreter tool

B The web_search tool

C The image_generation tool

D The file_search tool with the vector store ID

Sample question 8

The follow-up answer ignores the first question. Which value completes the second call so the conversation keeps its context?

```
project = AIProjectClient(endpoint=endpoint,
 credential=DefaultAzureCredential())
client = project.get_openai_client()

 first = client.responses.create(
 model="policy-chat",  # deployment name
 input="What is our refund policy?")

 follow_up = client.responses.create(
 model="policy-chat",
 input="Does it apply to sale items?",
 _____)
```

A previous_response_id=first.id

B store=False

C instructions=first.output_text

D temperature=0

Fabrikam Retail

Case study

Situation

Fabrikam is building a customer-service agent in Microsoft Foundry. 3,000 product manuals and return policies are already indexed in Azure AI Search with per-region security filters. Order data comes from an in-house Python function; refunds go through the finance team's MCP server.

Requirements

- Reuse the existing index and its security filters
- Get live order status from the in-house function
- Separate returns and billing specialist agents
- A person approves every refund call
- Remember customer preferences across sessions
- Block releases below the groundedness target

Sample question 9

Which knowledge tool should the agent use to meet R1?

A File search, with the manuals re-uploaded to the agent's vector store

B Code interpreter with the manuals attached as files

C The Azure AI Search tool connected to the existing index

D Web search restricted to fabrikam.com

Sample question 10

Which TWO actions meet requirements 2 and 4?

Choose TWO. Each correct answer is a partial solution.

A Define get_order_status as a function tool; the app executes it and returns the output

B Give code interpreter a connection string to the order database

C Add “always confirm before issuing refunds” to the agent instructions

D Add the refund MCP server with approval required, and handle approval requests in the app

Sample question 11

A customer who asked last month to be contacted in Spanish returns today. What meets Requirement 5?

A Increase max output tokens so the full chat history fits in context

B Enable agent memory — a long-term memory store scoped to each customer

C Hard-code each customer's preferences into the system instructions

D Run a Global Batch job each night over all past conversations

Microsoft

Implement computer vision solutions



Implement computer vision solutions

What the exam measures

- Design and implement image- and video-generation solutions
- Design and implement multimodal understanding workflows
- Implement responsible AI for multimodal content

10–15%

of exam questions

Computer vision: key concepts

What the exam measures

Design and implement image- and video-generation solutions

Images The images generated are original

Video an asynchronous process—you submit a job, poll for status, and download the result

Remix use the remix method with the original video's ID

Design and implement multimodal understanding workflows

Vision deploy a multimodal generative AI model

Content Understanding based on the creation of an analyzer

Implement responsible AI for multimodal content

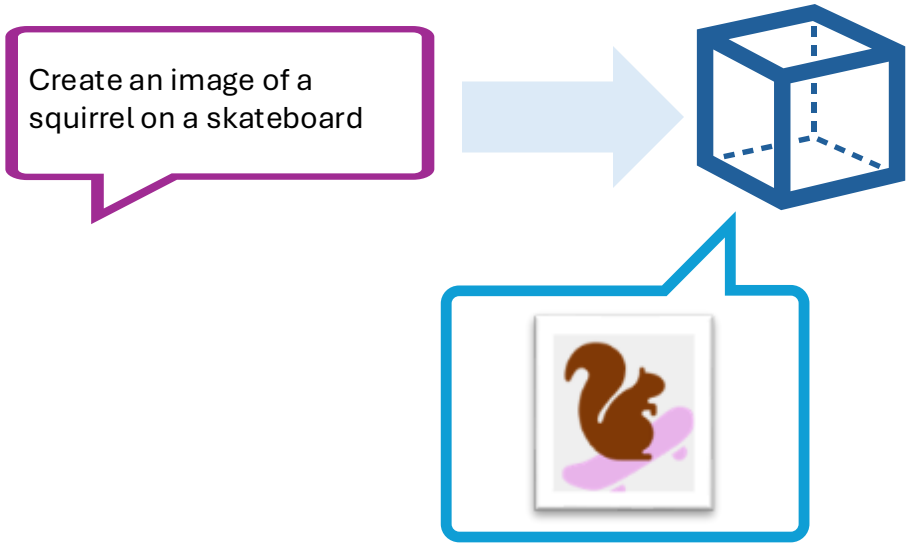
Reference images Reference images containing human faces are currently rejected.

Content filtering Prompts are subject to content moderation

Using an image-generation model

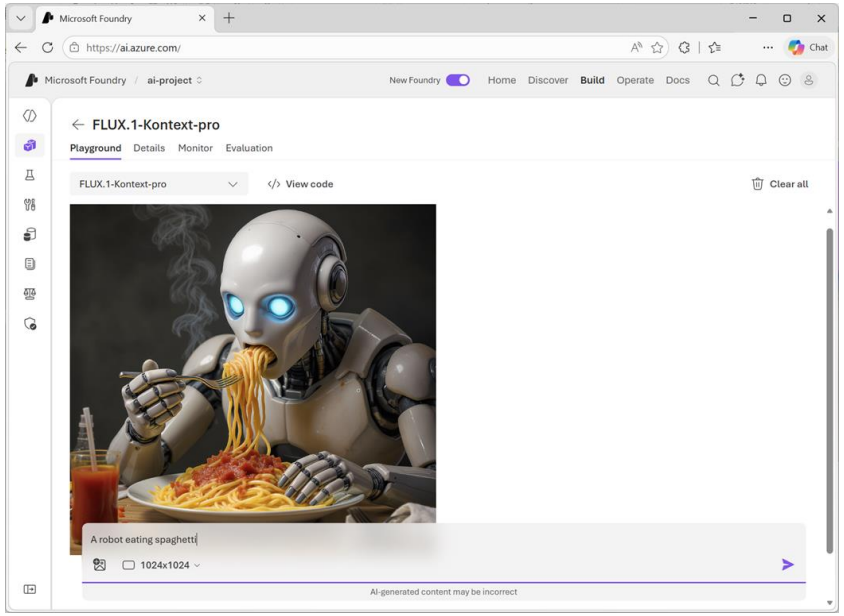
Deploy an image-generation model:

- GPT-Image
- FLUX.1-Kontext-pro
- …



Test the model in the playground

- Select an image size
- Attach a reference image
- (if supported and required)
- Enter a prompt describing the desired image



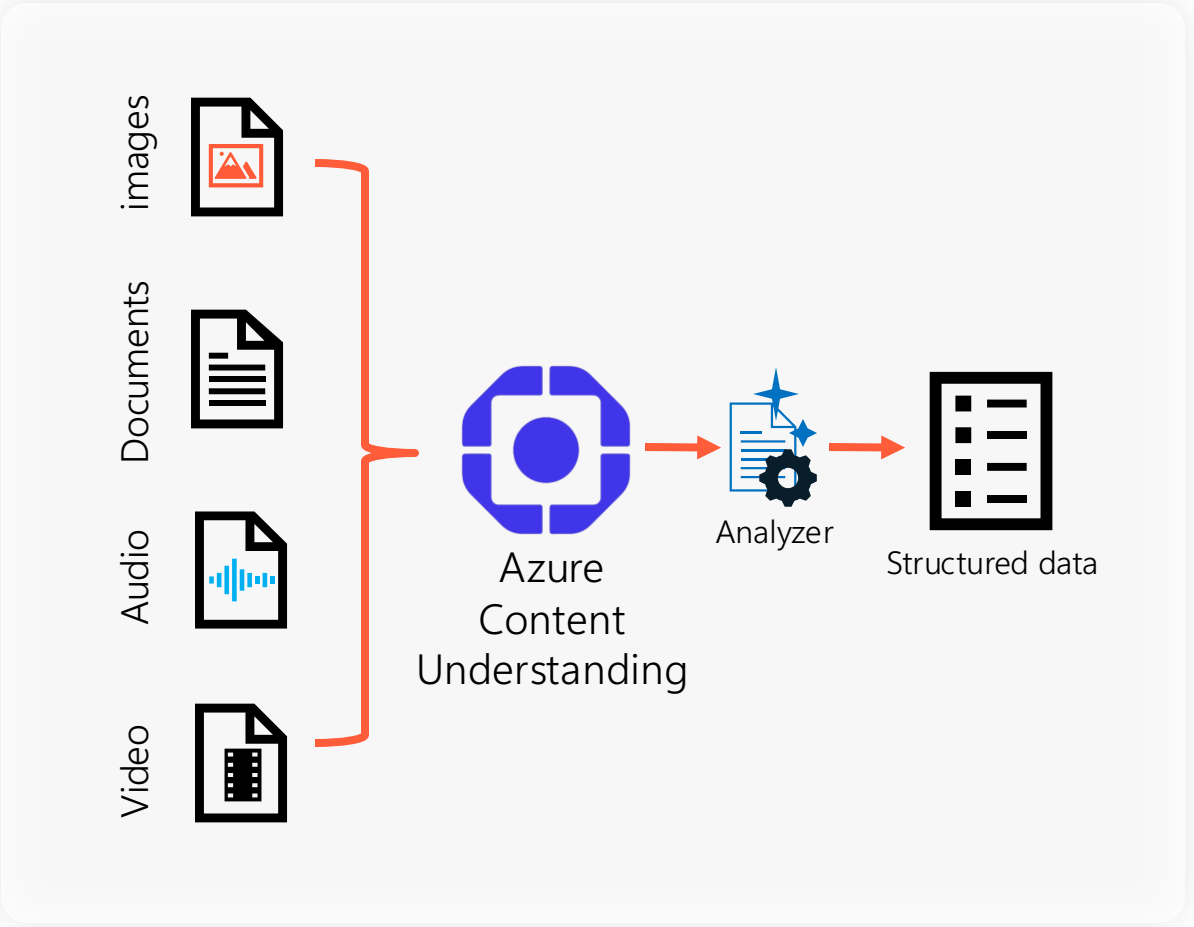
Azure Content Understanding in Foundry Tools?

Multimodal content analysis:

- Images
- Documents and forms
- Audio
- Video

AI-powered information extraction

- Pre-built and custom analyzers
- Single/Few-shot schema training



Filter unsafe multimodal content

- Implement filters to classify unsafe or disallowed visual content
- The same harm categories apply to images and generated media — violence, hate, sexual content, and self-harm
- Apply guardrails to image-generation and vision-capable model deployments in Microsoft Foundry
- Blocked prompts and completions return content-filter results that your application should handle gracefully

Responsible AI checkpoint

Map, measure, mitigate, and manage visual-content risks just as you do for text — guardrails are configured per deployment in the Foundry portal.

Sample question 12

The app starts a Sora 2 job and immediately tries to save the video, but the download fails. Which code should replace the blank line?

```
video = client.videos.create(
    model="sora-2", prompt=prompt,
    size="1280x720", seconds="4")

_____

content = client.videos.download_content(
    video.id, variant="video")
content.write_to_file("output.mp4")
```

A video = client.videos.remix(video_id=video.id, prompt=prompt)

B video.status = "completed"

C while video.status not in ["completed", "failed", "cancelled"]: time.sleep(20); video = client.videos.retrieve(video.id)

D video = client.videos.create(model="sora-2", prompt=prompt, seconds="8")

Sample question 13

A retailer wants to replace only the background of a product photo with a beach scene, keeping every product pixel unchanged. Which approach should you use?

A images.generate with a detailed prompt describing the product on a beach

B A vision-capable chat model asked to return the edited image

C A Content Understanding image analyzer with a “background” field

D An image edit request with the original photo, a mask covering the background, and a prompt

Microsoft

Implement text analysis solutions



Implement text analysis solutions

What the exam measures

- Apply language model text analysis
- Implement speech solutions

10–15%

of exam questions

Text analysis: key concepts

What the exam measures

Apply language model text analysis

Azure Language an API for common text analysis techniques

PII PII detection and redaction capabilities

Translator translate text and speech between languages

Implement speech solutions

Speech models transcribe speech to text · synthesize text to speech

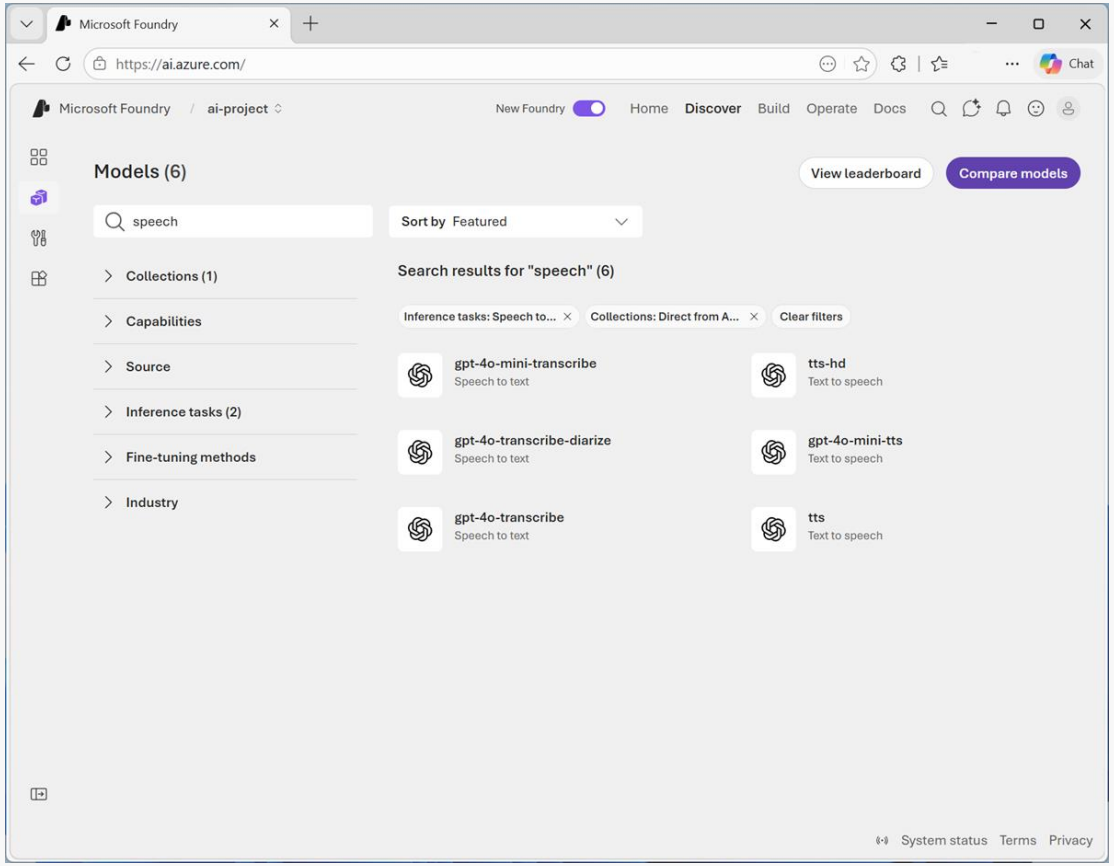
Speech translation translate spoken input and return the translation as text or speech

Voice Live real-time, bidirectional communication

Speech-capable models in Microsoft Foundry Models

Filter and search for:

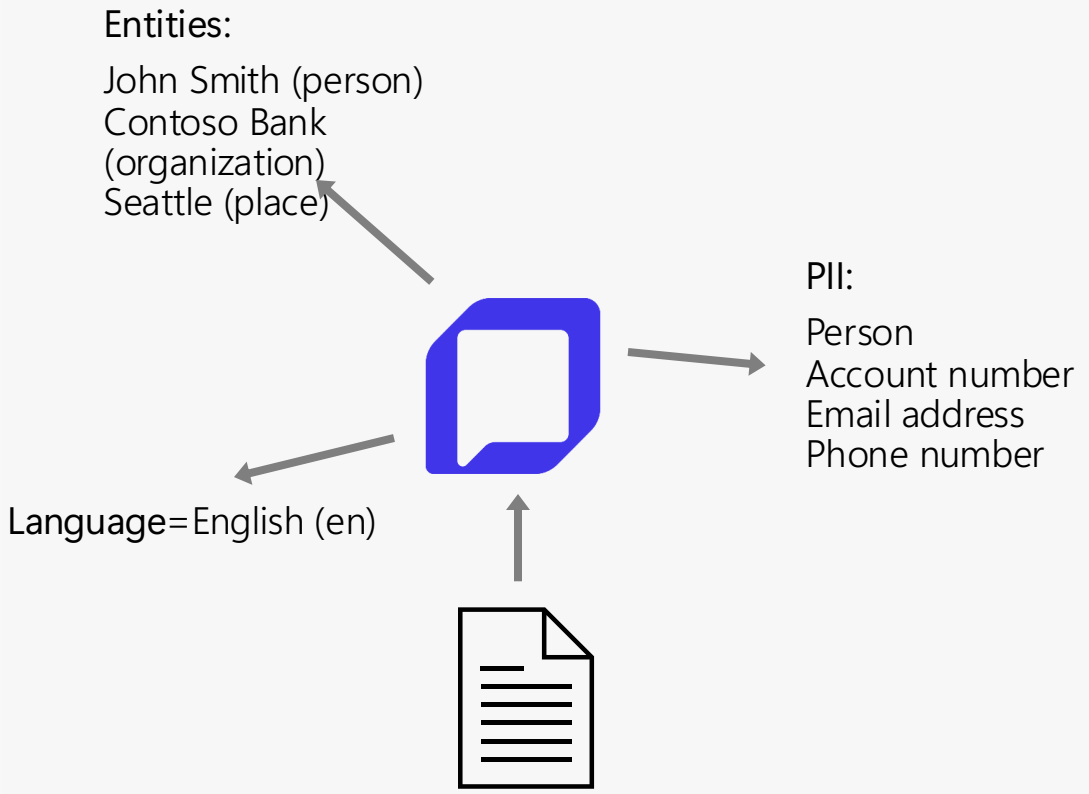
- Generative AI models that can transcribe speech to text
- Generative AI models that can synthesize text to speech



Azure Language in Foundry Tools

Prebuilt models for:

- Language detection
- Named entity recognition
- PII detection



Translation in Microsoft Foundry

Azure Translator in Foundry Tools

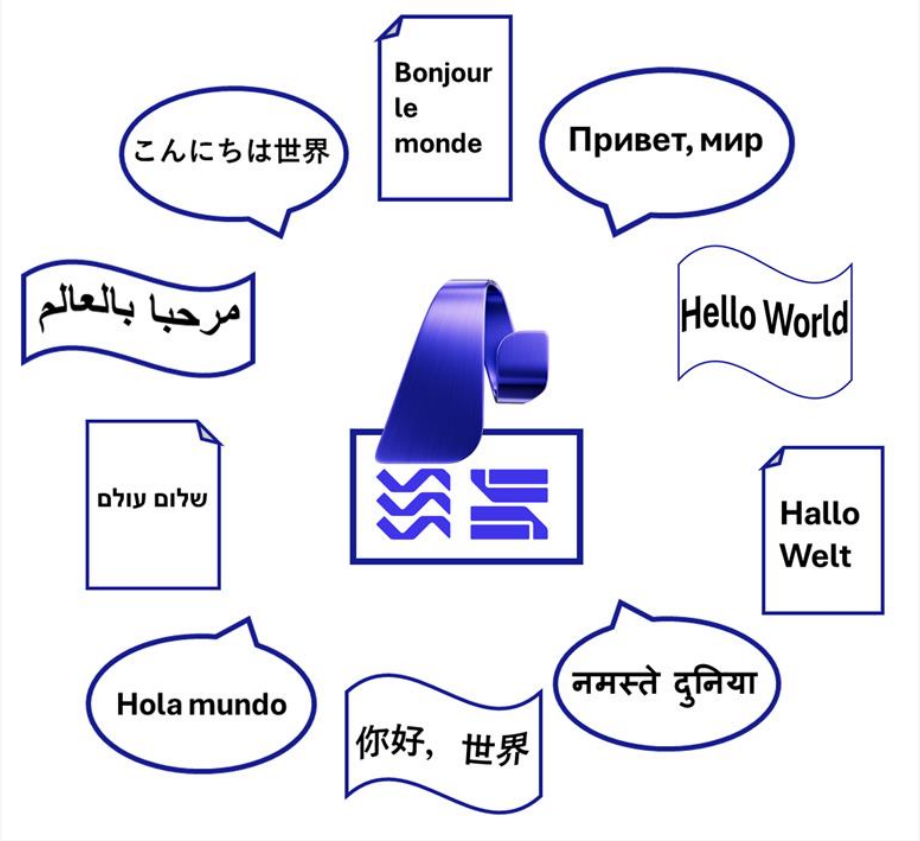
Multilingual translation for:

- Text
- Documents

Azure Speech in Foundry Tools

Multilingual speech translation:

- Speech to text
- Speech to speech



Sample question 14

Customer feedback must be shared with a partner after names, email addresses and phone numbers are masked, keeping the rest of the text readable. Which Azure Language call gives you the text to share?

A recognize_entities(), then replace each entity's text yourself

B detect_language(), sharing only documents detected as English

C recognize_pii_entities(), sharing each entity's category

D recognize_pii_entities(), sharing each document's redacted_text

Sample question 15

Complete the code so the app transcribes a French voice message from a WAV file.

```
speech_config = speech_sdk.SpeechConfig(
    subscription=key, endpoint=endpoint)
speech_config._____________ = "fr-FR"
audio_config = speech_sdk.audio.AudioConfig(
    filename="message.wav")
recognizer = speech_sdk.SpeechRecognizer(
    speech_config=speech_config, audio_config=audio_config)
result = recognizer.recognize_once_async().get()
```

A speech_synthesis_voice_name

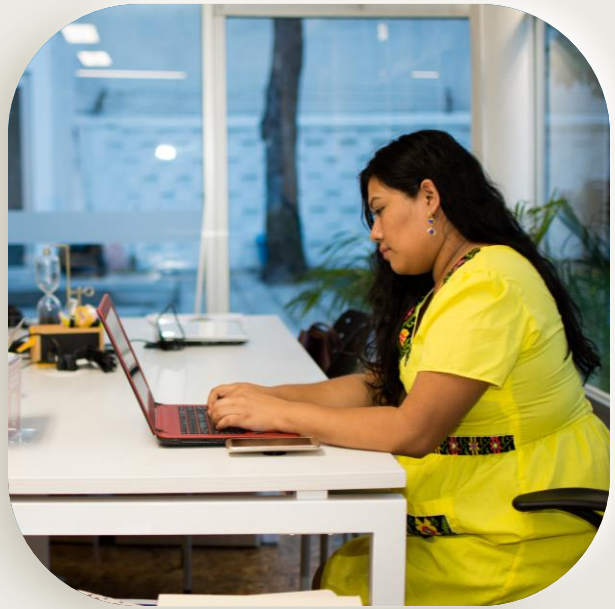
B speech_recognition_language

C add_target_language

D voice_name

Microsoft

Implement information extraction solutions



Implement information extraction solutions

What the exam measures

- Build retrieval and grounding pipelines
- Extract content from documents

10–15%

of exam questions

Information extraction: key concepts

What the exam measures

Build retrieval and grounding pipelines

Azure AI Search indexing and querying a wide range of data sources

Indexer an enrichment pipeline iteratively builds the documents

Knowledge store JSON objects, tables, or image files

Extract content from documents

Content Understanding extract insights and data from multiple kinds of content

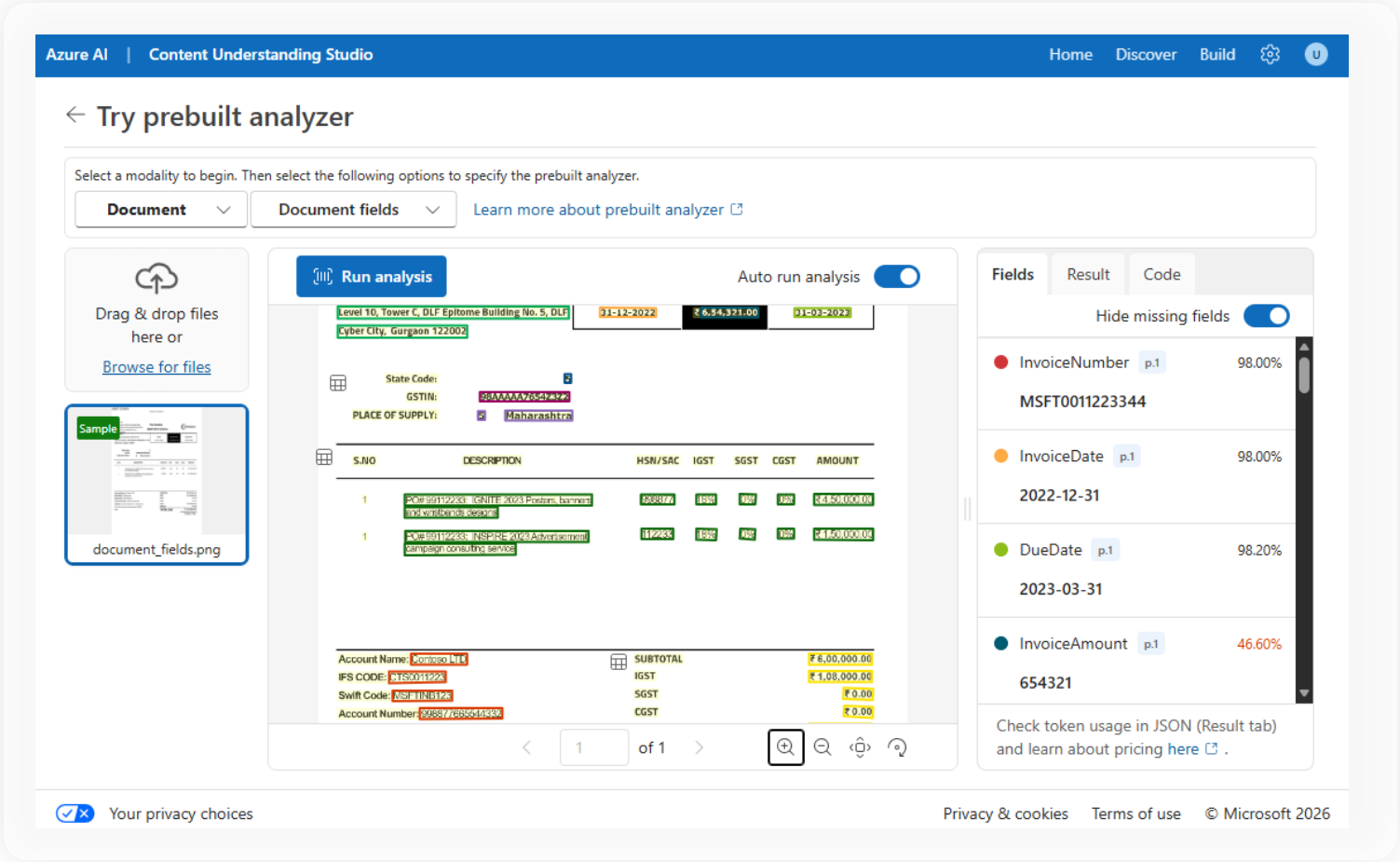
Document Intelligence uses OCR and deep learning models

Prebuilt models without any training required

Analyzing documents

Use Content Understanding to analyze documents and forms and retrieve specific field values

For example, extract key data values from an invoice to automate payment processing



Sample question 16

A search for comfortable hotel returns documents that contain either word, but users expect only documents that contain both. What should you change?

A Set queryType to full

B Add $orderby=rating desc

C Add select=title

D Set searchMode to All

Sample question 17

The first call to the Content Understanding API fails because the Foundry resource isn't ready. Which default model deployments does Content Understanding require?

A gpt-4o and text-embedding-ada-002

B GPT-4.1, GPT-4.1-mini and text-embedding-3-large

C Sora 2 and GPT-image-1

D None, because Content Understanding hosts its own models

Tips and tricks

1

Think in scenarios

Questions describe a business need — practice mapping it to the right service, model, deployment type, or API. Eliminate options that solve a different problem.

2

Read code with intent

Expect short Python snippets using the Foundry SDK and OpenAI SDK. You won't write code — you'll spot the right endpoint, parameter, or configuration.

3

Hands-on beats re-reading

Redo the labs — portal navigation, SDK patterns, and configuration steps are tested.

4

Manage the clock

Answer what you know first, flag the rest, and return. Every question scores the same — don't sink minutes into one scenario or one doc lookup.

Final checklist

From tonight to exam day

T–7 days

Practice and patch gaps

Full practice run above 80% in every domain; redo the labs for your weakest module.

T–1 day

Light review only

Key comparisons, guardrail categories, deployment types. Verify your Pearson VUE appointment and ID. Sleep.

Exam day

Set up early

Online: run the system test, clear your desk, stable connection. Test center: arrive 30 minutes early with ID.

In exam

Execute the strategy

Scenario first, options second. Flag and move on. Use Microsoft Learn sparingly. Answer everything.

Microsoft

Thank you